Interactive Digital Museum

The Voices of History

Gemini Live Agent Challenge 2026

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1. Project Description

The **Voices of History** is an interactive digital museum AI-powered web application that allows users to communicate with historical characters inside a virtual museum environment. The system combines conversational AI, voice interaction, visual animations, and computer vision to create an immersive educational experience.

Users can speak or type to famous personalities such as:

- **Tour Guide** (default virtual character)
  - Karim (tour guide)
- **Historical Characters**
  - Tut Ankh Amun
  - Nefertiti
- **Scientific Characters**
  - Isaac Newton
- **Artistic Characters**
  - Leonardo da Vinci

Each character responds using a personalized prompt that reflects their historical identity and communication style.

The system also supports **image recognition**, allowing users to upload an image of an antiquities, archaeological sites, and tourist attractions, or object. The AI analyzes the image and provides a description, simulating a museum guide explaining the item.

The application is implemented as a **real-time multimodal AI interface** using large language models, speech technologies, and visual processing to replicate a digital museum tour.

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2. Problem

Traditional museums face several limitations:

1. **Limited interaction**
   Visitors usually read static descriptions instead of engaging with historical figures.
2. **Language barriers**
   Not all visitors can understand the language used in museum explanations.
3. **Lack of personalization**
   Museum tours typically follow a fixed structure and cannot adapt to individual visitor interests.
4. **Limited accessibility**
   People who cannot physically visit museums miss the educational experience.
5. **Passive learning**
   Visitors consume information rather than interacting with it.

The Interactive Digital Museum addresses these issues by creating an **AI-driven interactive guide** that simulates conversations with historical figures and dynamically explains artifacts.

3. Key Features

3.1. AI Historical Character Conversations

Users can interact with multiple historical characters, each with a unique personality prompt and behavior. For examples:

- Tutankhamun explains ancient Egyptian artifacts.
- Isaac Newton discusses scientific discoveries.
- Leonardo da Vinci speaks about art and engineering.

3.2. Multimodal Interaction

The system supports three input methods:

- Text input
- Voice input
- Image upload

This allows users to interact naturally with the museum, then response with text and voice and lip movement animation.

3.3. Voice-Based Conversations

The system converts AI responses into speech using neural text-to-speech models, creating a more immersive museum guide experience.

3.4. Image Recognition for Artifacts

Users can upload an image of an object or artifact.
The AI identifies the item and provides a historical explanation.

3.5. Real-Time Streaming Responses

The AI response is streamed token by token, enabling:

- Faster interaction
- Real-time voice generation
- Natural conversation flow
- Interruptible

3.6. Dynamic Character Switching

Users can switch between different historical personalities at any time. The AI automatically adapts its behavior and knowledge based on the selected character, and also background place animation simulating its environment.

3.7. Interactive Visual Experience

The interface includes:

- Animated character avatars Idle and Talking animations
- Immersive background videos for each historical environment

3.8. Multilingual Support

The system supports both:

- English
- Arabic

Voice recognition and responses adapt automatically to the selected language.

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4. Key Tools and Techniques

The project integrates multiple AI and software technologies.

4.1. AI and Machine Learning

The application uses Google Gemini Models for:

- conversational AI Agent building
- image understanding and description
- streaming responses

Models used:

- `gemini-3.1-flash-lite-preview` for conversation and return streaming output
- `gemini-2.5-flash` for image recognition

4.2. Function Calling

The system uses **AI tool calling** to allow the model to trigger program functions such as:

- switching characters automatically

4.3. Speech Recognition Technologies

Voice input is converted into text using the Python library Speech-Recognition. This allows the user to interact with the system using natural speech, and it works perfectly in Egyptian Arabic STT.

4.4. Text-to-Speech

Speech responses are generated using Edge TTS. This produces realistic different voices for different characters, which is compatible with their gender, ages and persona.

4.5. Computer Vision

The system uses OpenCV and NumPy for image processing to:

- process uploaded images
- frame artifacts inside a museum-style golden frame
- display them visually in the interface

4.6. Web Application Framework

The application interface is built using Streamlit to provides:

- Chat interface
- Sidebar controls
- Real-time updates
- Multimedia display

4.7. Async Programming

The system uses **asynchronous programming with asyncio** to handle:

- AI streaming responses
- speech synthesis
- audio playback

This ensures smooth real-time interaction.

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5. Process and Functionality

The system follows several stages during user interaction.

5.1. User Interaction

The user can interact with the system through text input, voice recording, and image upload.

5.2. Input Processing

Depending on the input type: The text is sent directly to the Gemini AI agent. The voice recording is converted into text using speech recognition model then sent to Gemini AI agent. The image is processed and sent to the Gemini model for description.

5.3. AI Processing

The AI receives the user query, the selected character prompt, conversation history automatically made in the Agent session. The model generates a stream response consistent with the chosen historical personality.

5.4. Streaming Response

Instead of waiting for the full response, the system streams partial responses in real time. This allows the application to display text gradually, and generate speech while the response is being produced. So, the character will say “Okay let me thinking about that” and wait for the time duration for it that will takes only 2 seconds, during these time the first completed sentence will be finished and played, while playing the first sentence the second one will be finished, that makes the conversation works in real-time without any latency.

5.5. Speech Generation

Each completed sentence produced by the Gemini Agent AI is converted into speech using neural TTS with special voice for each character according to its gender, age, and persona. While the audio playback is synchronized with the streamed text.

5.6. Visual Character Animation

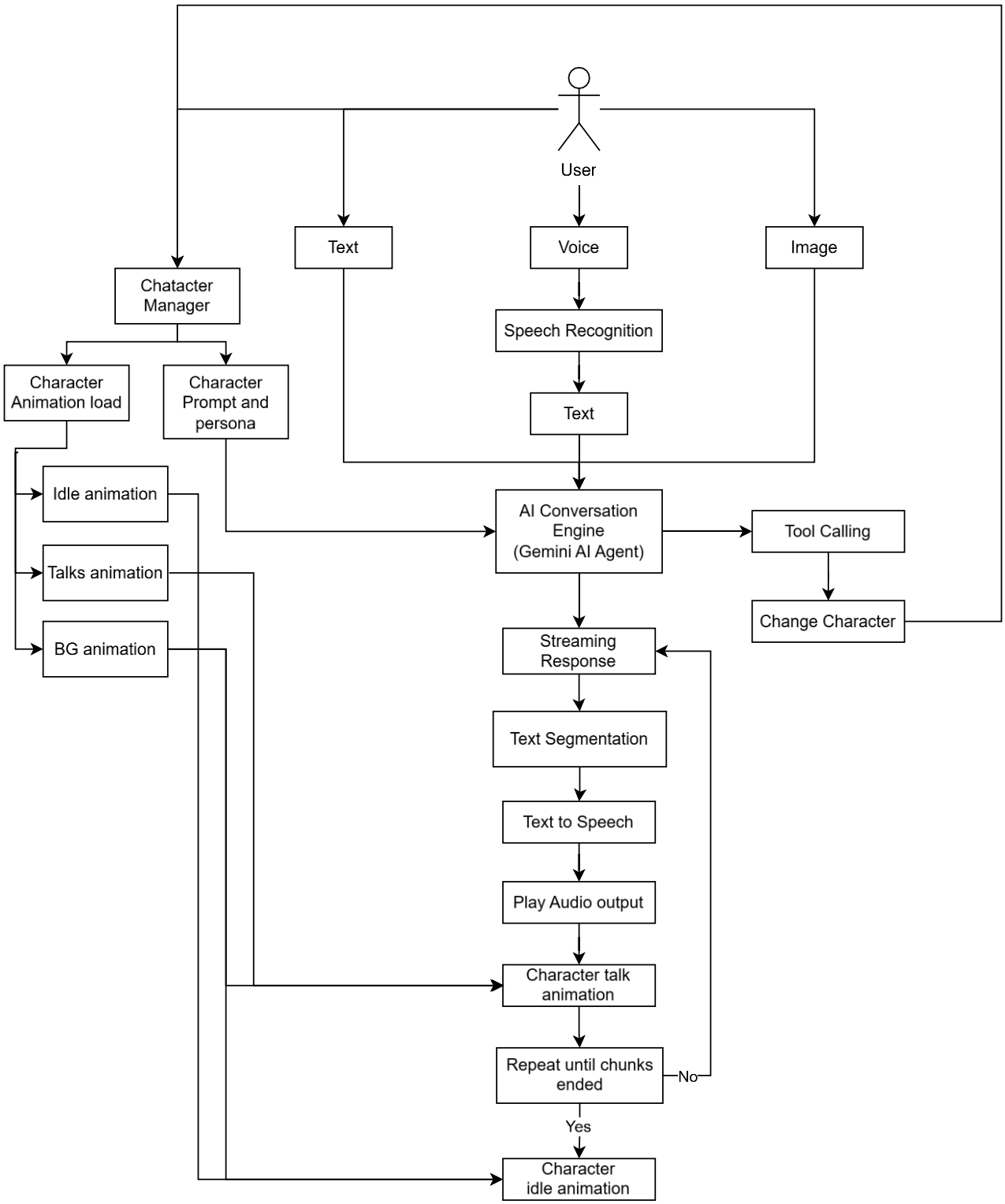
Two animation states exist for each character Idle animation which appears the character breathing, moving its head and eyes soft moves. And the talking animation which moves its lips to show as it is talking. When the character speaks, the talking animation is displayed, when not the idle animation is displayed. And a special background animated video for each character simulates its environment.

5.7. Artifact Image Explanation

When a user uploads an image:

1. The image is framed in a museum-style frame.
2. The Gemini AI model analyzes the image and returns the response in stream.
3. The system generates a descriptive explanation.
4. The explanation is spoken by the selected character.

Process Diagram



6. Findings and Learnings

While developing this project, several important insights are discovered as I have learned:

- **Streaming responses** make AI interaction feel faster and more natural compared to waiting for the full response.
- One of the main challenges was **displaying** streaming text, speech generation, and animation states using Streamlit **on parallel**. Achieving smooth synchronization required careful timing and asynchronous programming.
- **Combining** language models, speech technologies, and computer vision models creates a much richer interactive experience than text-based systems alone.
- **Working with asynchronous programming**, these is the first time I have used it ever.
- **Auto calling functions using Gemini** models allows the AI to takes program actions, such as switching characters, enabling more intelligent system behavior.
- **Not all problems solved with AI**, In my case I tried to use lip sync models but the latency was very high, and I need it to be in real-time so I decided to generate a design for each character and generate a design to each character background using Gemini models for one time, then make them idle with soft moves and breathing,… then apply lib sync only one time for each character, and moves the background images to be interacted with character the environment. Finally shifts between the stored idle character video and character talks video in fast way, And the human brain will do the other part and connect with the voice output and the libs moving to looks traditional.

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7. Future Improvements

Some future improvements that I haven't had time to add to the project due to time constraints

- Using Google Vertex AI RAG Engine to adding documents for the complete life story for each character and store them in Vector database then use it as API. So, the Gemini model will receive the user question and the needed data these will prevent hallucination 100%.
- Create a complete backend and database to the project which will make me saves the chat history to the user for future use if close the page and open it in other time.
- Create the web app using native web techniques as using Streamlit makes many obstacles as it prevent using Java Script code.

Conclusion

The Interactive Digital Museum (**Voices of History**) demonstrates how modern AI technologies can transform educational and entertainment experiences. By integrating conversational AI, voice interaction, and visual recognition, the project creates an engaging environment where users can interact with historical figures and explore artifacts in a natural and immersive way.

The project showcases the potential of **multimodal AI systems** to build interactive learning and entertainment platforms that go beyond traditional museum experiences.